

Aarhus University

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Game Assets

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Contents

1. Introduction 1

2. Design 2

3. Implementation 3

4. Testing Execution 7

5. Asset List 8

6. Conclusion 10

Bibliography 11

1. Introduction

Author: Kasper Stecher

The purpose of this appendix is to document the game assets used throughout the development of Parallax. Assets were utilized to support gameplay mechanics, environmental storytelling, visual readability, player feedback, and overall atmosphere.

The project combines a variety of asset types, including:

- 3D models
- textures and materials
- audio assets

Assets originated from both self-created content and third-party resources such as Unity's asset store **[1]** and Elevenlabs **[2]**. External assets were modified when necessary using tools such as Audacity **[3]** and Blender **[4]** to better fit the gameplay and visual style of the project.

2. Design

The asset design process focused on supporting the asymmetric gameplay experience between the Human and Cat roles while maintaining visual clarity during cooperative gameplay.

Environmental assets were selected and designed to:

- guide player movement
- improve puzzle readability
- create recognizable traversal paths
- support level progression

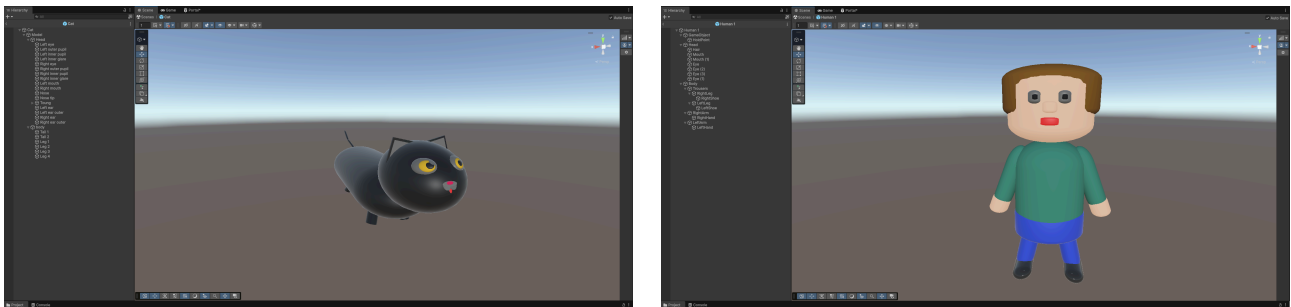


Figure 1: Playable Cat and Human character assets used in Parallax.

Character assets were designed to visually distinguish the two playable roles and reinforce gameplay differences between mobility-focused and interaction-focused mechanics.

Visual effects and lighting were designed to improve gameplay readability. This was especially important for mechanics such as:

- interaction highlighting
- activated objects
- cat vision
- environmental feedback

Audio assets were also considered part of the overall game design. Sound effects and music were used to:

- reinforce player actions
- provide environmental atmosphere
- improve interaction feedback
- strengthen character identity

3. Implementation

Game assets were integrated into Unity using a modular asset pipeline. Most imported assets required additional configuration before use in gameplay systems.

Implementation tasks included:

- converting materials to URP-compatible shaders
- configuring colliders
- adjusting object scaling
- setting up prefabs
- configuring lighting and shadows
- optimizing imported meshes

Many gameplay objects were implemented as reusable prefabs, including:

- doors
- buttons
- moving platforms
- lights
- puzzle objects

Audio assets were integrated into the custom audio system used throughout the project. Sounds were connected to gameplay systems such as:

- movement
- jumping
- interaction
- dialogue
- environmental events

Custom visual effects and shader-based materials were implemented using Unity Shader Graph within the Universal Render Pipeline (URP). Shader Graph was used to create stylized gameplay effects and animated materials while allowing effects to remain modular and adjustable during development.

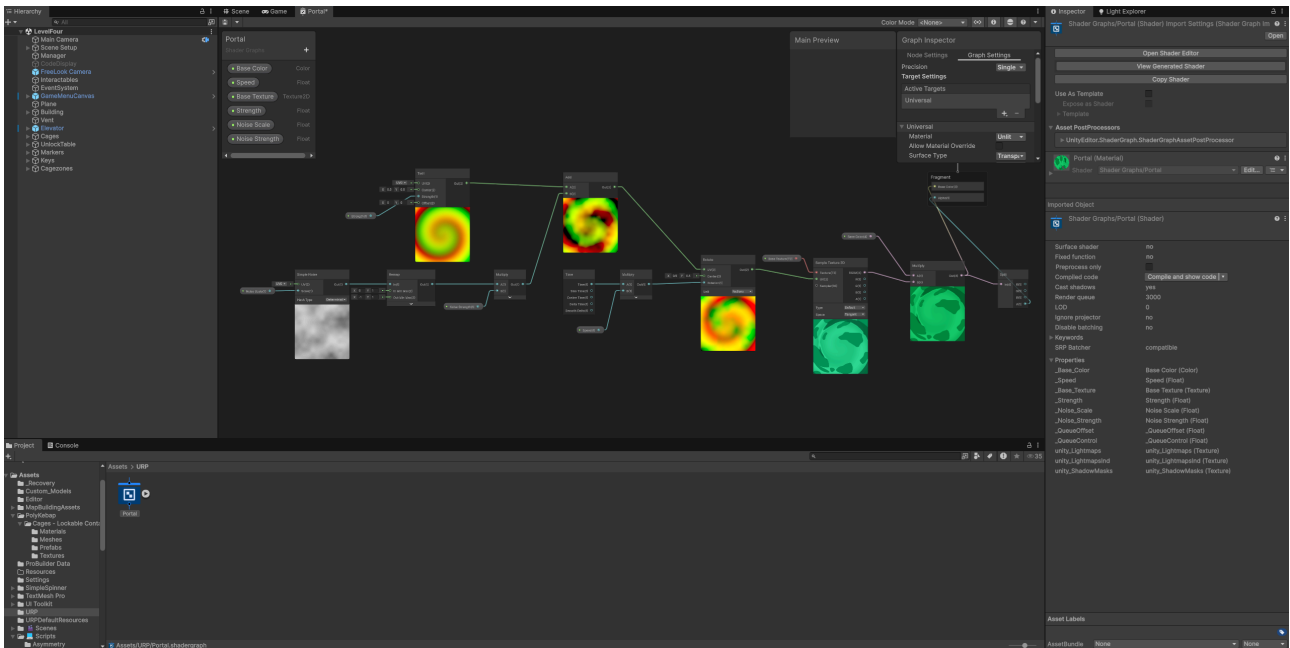


Figure 2: Example of a custom visual effect created using Unity Shader Graph [5] within the Universal Render Pipeline.

Visual effects were integrated into player systems and interactable objects to improve gameplay responsiveness and visual feedback.

Custom assets and audio processing tools were also utilized during development. Blender was used for creating and modifying simple 3D models, prototype geometry, and gameplay-related objects before exporting them into Unity.

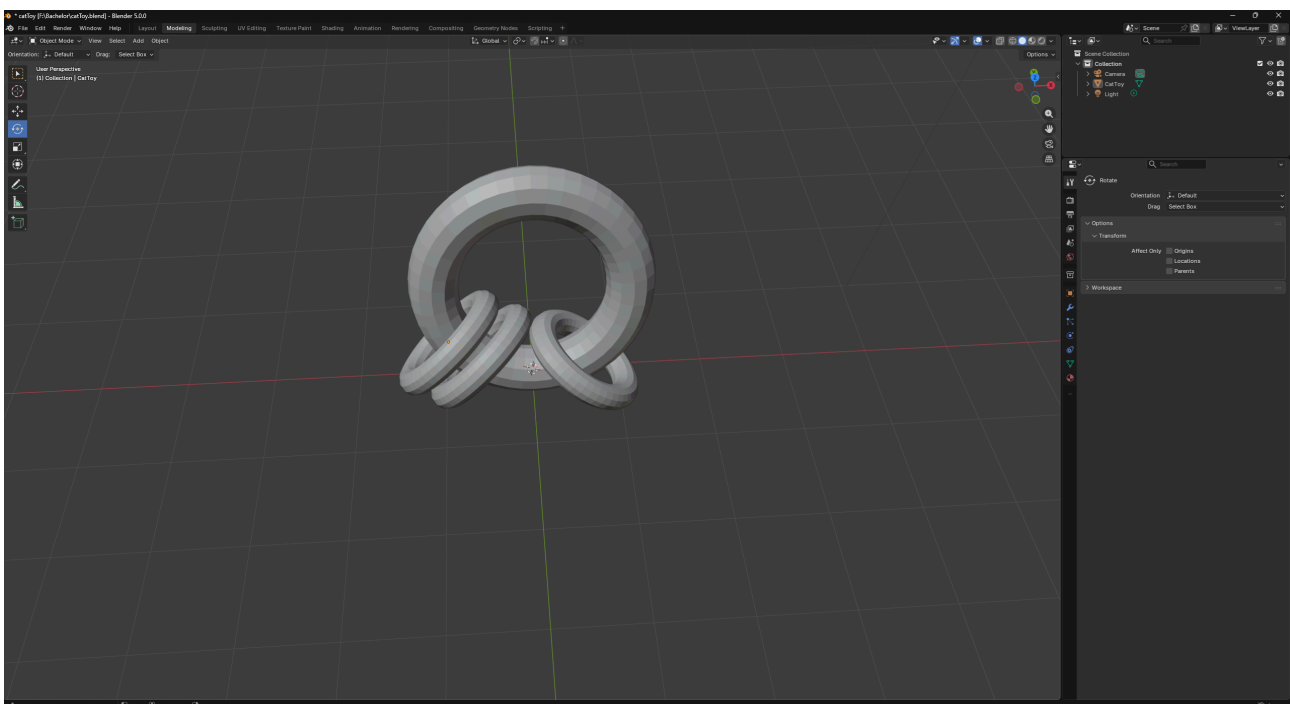


Figure 3: Example of a simple gameplay-related asset created and modified in Blender before integration into Unity.

Audio assets and dialogue recordings were processed using Audacity. This included trimming, adjusting volume levels, cleaning recordings, and preparing audio clips for integration into the gameplay audio system.

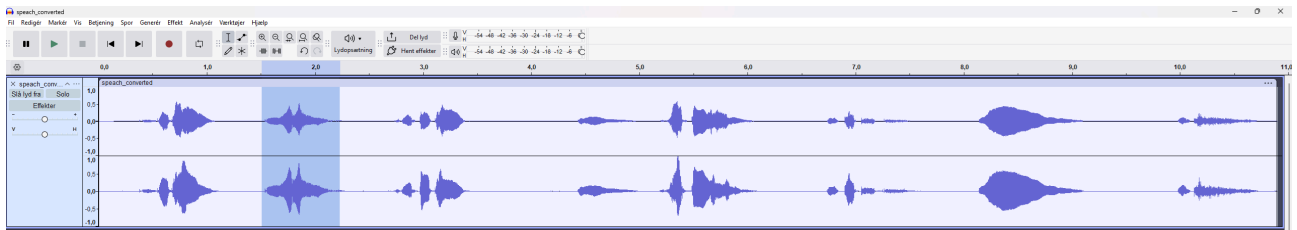


Figure 4: Example of audio processing and editing performed in Audacity before implementation into the game audio system.

Custom environment geometry was also created and modified in Blender to support gameplay requirements and level layout design. Simple modular structures such as walls, openings, and traversal spaces were prototyped externally before being integrated into Unity.

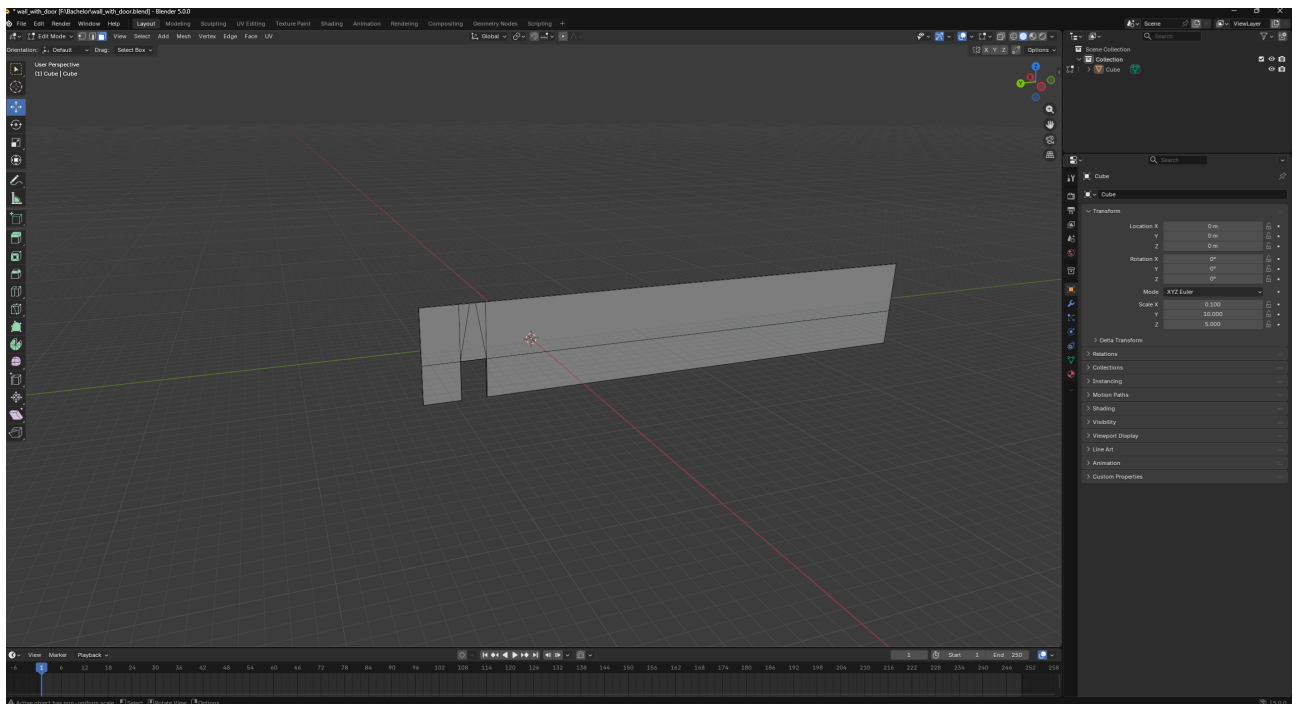


Figure 5: Example of custom environment geometry created in Blender for use as modular level assets within the game world.

The final game environments combine imported assets, custom geometry, lighting, shaders, and gameplay systems into cohesive gameplay spaces designed to support traversal, interaction, and cooperative gameplay.



Figure 6: Example of a completed gameplay environment combining imported and custom-made game assets within Parallax.

4. Testing Execution

Asset testing focused on ensuring gameplay readability, performance stability, and visual consistency across different gameplay scenarios.

Testing was performed both during development and through repeated gameplay sessions to evaluate asset readability and functionality during cooperative gameplay.

Testing included:

- validating collision behavior
- checking lighting visibility
- verifying audio playback
- evaluating gameplay readability
- identifying performance issues

Special attention was given to multiplayer synchronization, ensuring that:

- visual effects
- interactable states
- audio feedback

were displayed consistently across connected clients.

Several issues were identified during testing, including:

- material compatibility problems after URP conversion
- inconsistent object scaling
- lighting inconsistencies
- collider issues on imported models

These issues were gradually corrected throughout development.

5. Asset List

Level assets

- <https://assetstore.unity.com/packages/2d/textures-materials/yughues-free-architectural-materials-13234>
- <https://assetstore.unity.com/packages/3d/props/lockable-containers-cages-347612>
- <https://assetstore.unity.com/publishers/26314>
- <https://assetstore.unity.com/publishers/37259>
- <https://assetstore.unity.com/packages/2d/textures-materials/nature/handpainted-grass-ground-textures-free-top-down-rpg-terrain-tool-187634>
- <https://assetstore.unity.com/packages/2d/textures-materials/nature/yughues-free-sand-materials-12964>
- <https://assetstore.unity.com/packages/3d/vegetation/trees/polygon-trees-224068>
- <https://assetstore.unity.com/packages/3d/vegetation/trees/realistic-tree-9-rainbow-tree-54622>
- <https://assetstore.unity.com/packages/2d/textures-materials/floors/realistic-terrain-textures-free-279940>
- <https://assetstore.unity.com/packages/3d/props/electrical-shield-118266>
- <https://assetstore.unity.com/packages/3d/props/interior/free-sockets-and-switches-233085>
- <https://assetstore.unity.com/packages/3d/props/furniture/low-poly-simple-furniture-free-240197>
- <https://assetstore.unity.com/packages/3d/props/p3d-indoor-design-starter-kit-3d-models-furniture-264116>
- <https://assetstore.unity.com/packages/3d/vegetation/environment-pack-free-forest-sample-168396>
- <https://assetstore.unity.com/packages/3d/props/industrial/industrial-props-kit-84745>
- <https://assetstore.unity.com/packages/3d/props/tools/simple-keys-231162>

UI and Menu assets

- <https://opengameart.org/content/ui-pack>
- https://www.spritters-resource.com/pc_computer/recroom/asset/178300/
- <https://www.svgrepo.com/svg/528877/box>
- <https://iconmonstr.com/eye-3-svg/>
- <https://www.svgrepo.com/svg/532244/gear>
- <https://www.svgrepo.com/svg/510850/book-open>
- <https://iconmonstr.com/key-13-svg/>

- <https://www.svgrepo.com/svg/174286/pc-mouse-top-view>
- <https://www.svgrepo.com/svg/521789/play>
- <https://www.svgrepo.com/svg/499637/resume>
- <https://www.svgrepo.com/svg/311202/sign-out>
- <https://www.svgrepo.com/svg/522667/sound-off>
- <https://www.svgrepo.com/svg/522670/sound-volume-2>
- <https://www.svgrepo.com/svg/341308/voice-activate>
- <https://assetstore.unity.com/packages/2d/gui/icons/simple-spinner-progress-indicators-for-ui-237500>

The icons used for the Private Lobby, Public Lobby, and Guide features were originally sourced from svgrepo. During the final documentation phase of the project, the original assets could no longer be located on the platform. Searches using the icon names, categories, and visual characteristics did not return the exact files previously used in the application. It is therefore assumed that the assets have either been removed or are no longer publicly available on svgrepo. Consequently, direct source links to the original icon assets cannot be provided in this report.

6. Conclusion

The use of both custom and third-party assets allowed the project to maintain a functional and visually readable game environment while prioritizing gameplay system development. Assets played an important role in supporting the overall feel of the game by improving cooperative gameplay, environmental communication, player feedback, atmosphere, and gameplay readability. The modular implementation approach also allowed assets to remain reusable and easier to maintain throughout development.

Future improvements could include the creation of additional custom-made assets, improved visual consistency across environments and gameplay objects, expanded animation sets, more advanced visual effects, additional audio variation, and further optimization of imported resources.

Bibliography

- [1] U. Technologies, "Unity Asset Store." [Online]. Available: <https://assetstore.unity.com/>
- [2] ElevenLabs, "ElevenLabs." [Online]. Available: <https://elevenlabs.io/>
- [3] A. Team, "Audacity." [Online]. Available: <https://www.audacityteam.org/>
- [4] B. Foundation, "Blender." [Online]. Available: <https://www.blender.org/>
- [5] D. Ilett, "10 Shaders in 10 Minutes - Unity Shader Graph." [Online]. Available: <https://www.youtube.com/watch?v=vje0x1BNpp8>